

Coherence Is Not a Green Light: The Limits of Self-Consistency as a Label-Free Reliability Proxy for Medical Vision–Language Models

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Abstract. Medical Vision Language Models (VLMs) are deployed faster than reference labels can audit them - especially in the self-hosted setting an institution must audit without vendor labels - motivating *label-free* proxies for report trustworthiness. The most intuitive is *self-coherence*: accept a report when its yes/no conclusion follows from the radiographic evidence the report itself describes. We test whether self-coherence is a usable acceptance gate for chest-radiograph finding detection across three architecturally distinct open-weight medical VLMs evaluated zero-shot, with coherence scored by a ground-truth-blind LLM judge whose criteria were signed off by a hepatologist and whose labels meet a pre-registered agreement threshold at the point estimate (Gwet's AC1). Framed as a one-sided bound rather than a naked null, the probability that a coherent report is *correct* sits far below any clinically usable threshold - even at the optimistic bound of its confidence interval - so no “accept-when-coherent” screen is deployable among the open-weight models we evaluate. Coherence is not zero-information (coherence→correctness AUROC ≈ 0.56), so the honest claim is directional: coherence fails as a *green light*, while incoherence retains weak value as a *red flag*. Most tellingly, forcing a fabricated measurement causally *decouples* coherence from correctness: on one architecture it collapses coherent-correct predictions (53%→27%), on another it inflates coherence without improving accuracy—so the metric is manipulable, not reliable. We frame this as a *negative result with a causal stress test* - not a proposed replacement gate - cautioning label-free deployment: self-inspection catches the model that contradicts itself and passes the far larger population that is confidently, coherently, and completely wrong.

Keywords: Medical vision language models · Reliability · Hallucination · Self-consistency · Chest radiography · Uncertainty.

1 Introduction

A radiology VLM in deployment produces reports faster than any institution can label them. The operational question is therefore not “what is this model's population AUROC” (which needs a reference standard) but “should I trust *this* report, now, without ground truth.” *Self-coherence* is the intuitive answer: read the model's own description of the image and check whether its final yes/no conclusion follows from the signs it claims to see. A report stating “the cardiac silhouette is enlarged, therefore cardiomegaly” is coherent; one stating “the heart is normal in size, therefore cardiomegaly” is not. The hope is that coherence tracks correctness closely enough to serve as a deployable acceptance gate. The same intuition underlies self-consistency checks that resample a model and trust unanimous answers. We test the *single-report* form - rationale→conclusion coherence under greedy decoding - and scope out sampling-based self-consistency, a related but distinct signal we do not evaluate here.

We test this hope directly on chest-radiograph finding detection and find it unsound. Our aim is *falsification, not replacement*: we make no claim to a better label-free gate, and evaluate coherence on the terms its proponents assume. Our contributions:

1. **Coherence is not a deployable acceptance gate**, established as a *one-sided bound* with confidence intervals rather than a point-null: the probability a coherent report is correct is far below any usable clinical threshold even at its optimistic bound. We report the (small but nonzero) coherence→correctness discrimination ourselves, so the claim is directional - coherence fails as a *green light*, incoherence remains a weak *red flag* (Sec. 5.2).
2. **A causal demonstration**: forcing a model to state a fabricated measurement (cardiothoracic ratio) *decouples* coherence from correctness—collapsing coherent-correct predictions on one architecture while inflating coherence without improving accuracy on another - so coherence is *manipulable* independently of correctness (Sec. 5.3).

We present cross-architecture *evidence* from three representative open-weight medical VLMs spanning distinct archetypes, not a field-wide law.

2 Related Work

Hallucination and over-confidence in medical VLMs are widely documented [12, 10], and uncertainty estimation for safe deployment is the theme of this venue. A range of *label-free* trust signals have been proposed as deployable gates: model self-verification, in which a system is asked whether its own answer is correct [15]; semantic-clustering measures that flag generations whose *meaning* is unstable across samples [14, 13]; and coherence between a stated rationale and the final answer. LLM-as-judge has become a standard tool for scalable evaluation of free-text generations [11]. Reliability in radiology report generation has otherwise been quantified with *label-dependent* metrics - CheXbert-vector agreement and clinical entity/relation overlap [3, 16, 17] - which cannot score a single report at deployment time without a reference standard. We connect these threads by asking whether coherence, scored by a clinician-vetted, ground-truth blind judge, is a *sound* acceptance gate for medical VLMs, and answer in the negative with a one-sided-bounding lens that pre-empts over-claiming a null. That coherent reasoning can accompany an incorrect answer is itself documented in the general LLM hallucination literature; our contribution is the systematic, interval-based, and causally stress-tested evaluation of it as a label-free acceptance gate for medical VLMs.

3 Data and Models

Cohort. We use a frozen, patient-disjoint, SHA-locked cohort drawn from CheXpert Plus [2]: 5,000 studies from 3,525 patients, split 3,500 train / 1,500 held-out test at the patient level with zero leakage across splits or the five within-train cross-validation folds (iterative multilabel stratification [4]; train-test prevalence drift ≤ 0.10 pp). We report the competition-standard **CheXpert-5** [1] (Atelectasis, Cardiomegaly, Consolidation, Edema, Pleural Effusion); each finding clears a ≥ 100 -positive test reporting gate. Images are 8-bit min-max PNGs with no VOI-LUT windowing applied - a deliberate choice, as windowing-tag availability is significantly associated with five

of the seven cohort findings, so selective windowing would confound normalisation with finding identity. Ground-truth labels are standard impression-derived CheXbert labels [3].

Models (zero-shot, no fine-tuning). We evaluate three open-weight medical VLMs chosen as distinct archetypes (Table 1): a *pattern-matcher* (MedGemma-4B [7], SigLIP+Gemma-3), a *reasoner* (CheXOne [8], post-trained on Qwen2.5-VL-3B [9]), and a *degenerate/evasive* model (LLaVA-Med-7B [10], Mistral).

Table 1. The three medical VLMs, framed as behavioural archetypes.

Model	Backbone	Archetype / characteristic behaviour
MedGemma-4B	SigLIP + Gemma-3	Pattern-matcher: commits readily, estimates genuinely
CheXOne	Qwen2.5-VL-3B	Reasoner: hedges/abstains, anti-calibrated when it commits
LLaVA-Med-7B	Mistral (single-encoder)	Degenerate/evasive: prompt-invariant, narrates instructions

4 Method

Coherence elicitation. For each (image, finding) we prompt a four-step report: describe the image; focus on the relevant region; reason step by step about the finding; and on the last line write `FINAL: YES/NO`. Generation is greedy; ground truth is stored but never sent downstream. Each model was run under its official chat template and image-preprocessing pipeline.

Ground-truth blind judge. Coherence is scored by an LLM judge (Claude Opus 4.8) that sees only the model’s free-text report and asks a single question: does the `FINAL` answer follow from the radiographic evidence the report *itself* states, against explicit per-finding sign lists derived from the Fleischner Society glossary [5] and standard chest radiography references? This is *internal* consistency, not correctness - exactly the check a label-free deployment could run. We define a report as *committed* if a yes/no can be extracted (echoed instruction templates map to *uncertain* and are excluded), *coherent* per the judge, and *correct* if the answer matches the label. The judge scored every elicited report once—31,500 in total (10,500 per model: 7,500 on the five neutral findings and 3,000 across the intervention arms of Sections 5.3 and 5.4).

Judge validation. The judge is load-bearing, so it is validated in two pre-registered gates. *Gate A:* the per-finding sign lists are drawn from the Fleischner Society glossary [5] where applicable, and from standard chest-radiography criteria, and were confirmed by a hepatologist experienced in chest-radiograph interpretation as applying these textbook-standard criteria, then frozen as a single source of truth. This checks that the judge applies *published radiographic definitions* (e.g. “does an enlarged silhouette on a PA film support cardiomegaly”) - what a coherence check requires—rather than that its calls match expert reads; clinical ground truth comes from the CheXbert labels, not the reviewer. Neither clinician is a board-certified radiologist; we mitigate this by binding the rubric to the Fleischner (or published radiographic) definitions rather than to individual expert judgement, and note it as a limitation. *Gate B:* a held-out sample of 60 reports (20 per model) was labelled for coherence by an independent general practitioner, blind to the judge, on a three-option form (*coherent / incoherent / unclear*); the labels were then verified by the hepatologist and scored against the judge with three-

category Gwet's AC1 [6] (preferred over Cohen's κ under high agreement). Scoring was pre-registered as pooled (≥ 0.60); because the degenerate model contributes trivial *unclear-unclear* concordance that *inflates* the pooled coefficient (pooled AC1=0.79), we report the more conservative non-degenerate agreement (n=40) as primary, with per-model figures descriptive. The judge's *unclear* label is derived (from an unclear answer or ambiguous stated evidence), making the agreement a further conservative bound. Both pre-registered thresholds are met at the point estimate (Table 4); 95% intervals are percentile bootstrap (5,000 resamples), which the modest calibration sample widens—the AC1 lower bound (0.50) sits below the pre-registered 0.60, so the threshold is met on the point estimate but not across the whole interval, and a larger calibration set with a second, different-family judge is the pre-planned robustness check.

Defensible-null statistics. A “no useful signal” claim requires intervals. We report P(wrong|coherent) and P(correct|coherent) with 95% **patient-clustered** bootstrap intervals—resampling the 1,067 test patients (5,000 resamples) rather than treating the findings that co-occur within a patient as independent, which runs wider than iid Wilson but leaves every deployment-bar conclusion unchanged - together with the coherence→correctness AUROC and a one-sided non-inferiority verdict against a clinical acceptance bar (≥ 0.90 , an illustrative high-stakes reference; the appropriate bar is setting-dependent, so we also report against a lenient 0.70 line - which no group clears). Coherence is binary, so this AUROC is the single-operating-point ROC area — equivalently balanced accuracy - not a swept-threshold curve. To control contamination from a degenerate model whose coherence is canned by construction, we report a **non-degenerate primary** (pattern-matcher + reasoner) alongside the pooled figure, on committed, non-echoed responses. The base rate is the marginal P(wrong); the balanced comparison is what is interpretable, and deployment fractions will differ.

5 Results

Unless explicitly noted, all numbers below are on the sealed 1,500-study test split, accessed once after the analysis pipeline and clinician-signed judge were frozen (Section 4).

5.1 The models are unreliable

Even scored generously - on committed responses only, ignoring the reasoner's 32% and the degenerate model's 47% abstentions - 14 of 15 model×finding cells fail to beat an always-negative baseline (the lone exception is the pattern-matcher on the highest-prevalence finding). The models are not usefully accurate; this is the backdrop against which a label-free gate would have to work. This near-chance regime is deliberate, not a weakness: the deploy-before-labels setting we target is precisely where an institution cannot yet know a model is weak, so a label-free gate is most needed in exactly this regime; we do not claim it must also fail for stronger, better-grounded models — only that the causal result (Sec. 5.3), where manipulability is independent of accuracy, does not rely on the models being good.

5.2 Coherence is not a deployable acceptance gate

Table 2 and Fig. 1 give the core result. On the contamination-controlled primary (n=12,609 committed responses over 1,067 patients), a coherent report is correct with probability 0.623 (patient-clustered 95% CI [0.609, 0.637]); even the optimistic bound is far below a clinical acceptance threshold - and below even a lenient 0.70 bar - so no “accept-when-coherent” screen is deployable for the open-weight models we study. Coherence is *not* zero-information - coherence→correctness AUROC is 0.56, and the coherent wrong-rate (37.7%) sits 4 pp below the base rate (41.3%; interval excludes zero) - which is exactly why we make the directional claim: **coherence fails as a green light**. The asymmetry is the story: P(wrong|incoherent) is 22 pp higher than P(wrong|coherent) (59.6% vs. 37.7%), so **incoherence retains weak value as a red flag**, but a reject-signal is not verification. A coherence screen still misses 76% of all errors, because errors are overwhelmingly coherent: coherence runs 80–99% across the three models - wrong calls are self-consistent fabrications, not contradictions, which is *why* an internal-consistency check cannot catch them. The null holds for each architecture individually: P(correct|coherent) is 0.71 for the pattern-matcher (clustered CI [0.695, 0.730], straddling the lenient 0.70 line rather than clearing it), 0.48 for the reasoner, and 0.60 for the degenerate model. Most tellingly, the degenerate model is 99.5% coherent yet its coherence predicts correctness at chance (AUROC 0.498): **the most coherent model is precisely the one where coherence carries the least information** - the cleanest statement of the dissociation.

Table 2. Coherence as an acceptance gate (sealed 1,500-study test split, committed + non-echoed responses). **Non-degenerate** (pattern-matcher + reasoner) is the primary; pooled includes the degenerate model, whose near-universal coherence adds little because it barely moves the figure. No group nears a clinical acceptance bar; even the strongest single model (the pattern-matcher, 0.71) at best grazes a lenient 0.70 line. CIs are patient-clustered bootstrap (1,067 patients).

Group	P(corr coh) [95% CI]	base P(wr)	P(wr incoh)	AUROC
Non-degenerate (primary)	0.623 [0.609, 0.637]	0.41	0.60	0.56
Pooled (incl. degenerate)	0.615 [0.604, 0.627]	0.41	0.59	0.55

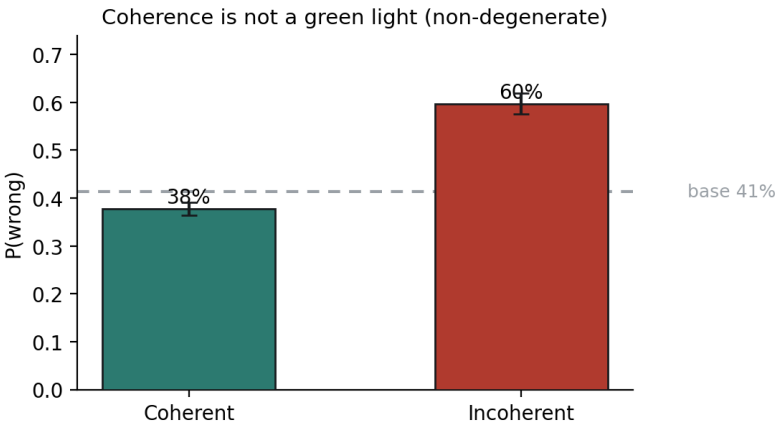


Fig. 1. Coherence is not a green light. Wrong-rate for coherent vs. incoherent reports against the marginal base rate, with 95% CIs (non-degenerate primary, n=12,609). Incoherence lifts the wrong-rate (59.6%, a red flag); the coherent wrong-rate (37.7%) sits only 4 pp below the base rate (41.3%) - far from the clinical acceptance line.

5.3 Coherence is manipulable: a causal intervention

If errors announced themselves as contradictions, a coherence gate would work. They do not. We stress-test this by forcing each model to commit to a fabricated cardiothoracic-ratio (CTR) value - a genuine radiographic quantity a clinician computes, so the demand is realistic, not a trick - and compare against the neutral prompt on the same studies (paired, committed-in-both). The fabricated number causally *moves* the coherence metric, in *opposite* directions by architecture, and in neither case toward correctness (Table 3, Fig. 2). On the pattern-matcher, coherence *falls* (77→63), but the coherent reports that remain are far worse: coherent-*correct* predictions collapse (53%→27%) while coherent-*wrong* rises (24%→37%), so among reports that stay coherent the error rate nearly doubles (31%→58%) - the invented measurement **converts correct-coherent reports into confident, coherent errors**. On the reasoner the same intervention does the opposite to the metric - coherence *rises* (63→80) - yet accuracy does not improve (error among coherent reports essentially flat, 35%→36%): it manufactures coherence without manufacturing correctness. The degenerate model is prompt-invariant (0% of answers change), neither breaking nor inflating. **A self-inspection layer watching for contradiction would, on one architecture, watch the fabrication smooth the report and, on the other, watch its correct-coherent predictions disappear** - in both cases blind to the injected error. We restrict the probe to Cardiomegaly because CTR is the one manipulation grounded in a real clinical measurement; “effusion volume as a decimal” and “consolidation as a percent” are not clinical constructs and would be dismissable as artificial tasks.

Table 3. Forced-CTR intervention (Cardiomegaly, paired, committed-in-both). Coherence moves oppositely by architecture; neither model converts it into correctness. Percentages; → reads neutral to forced-CTR.

Model (n)	Coherence	Coh-correct	Coh-wrong	Reading
MedGemma-4B (500)	77→63	53→27	24→37	collapses coherent-correct
CheXOne (186)	63→80	41→51	22→28	inflates coherence, no accuracy gain
LLaVA-Med-7B (278)	99→100	75→76	24→24	prompt-invariant (degenerate)

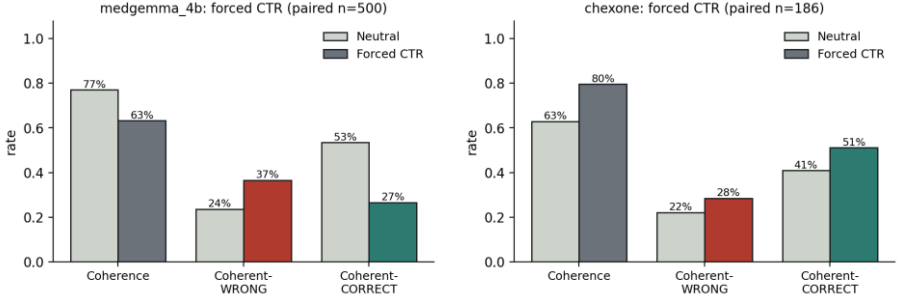


Fig. 2. A fabricated CTR decouples coherence from correctness, oppositely by architecture. Neutral vs. forced-CTR (Cardiomegaly, paired). *Left* (pattern-matcher, MedGemma): the fabrication collapses coherent-*correct* predictions (53%→27%) and lowers coherence. *Right* (reasoner, CheXOne): the same injection *inflates* coherence (63%→80%) without improving accuracy. Neither turns the manipulation into correctness.

5.4 Not remediable by prompting

A directed sign-check prompt does not remediate the failure. Its effect is finding-specific and *bidirectional*—it raises the pattern-matcher’s coherent-correct rate on its weakest finding (Atelectasis, 37%→42%) but is neutral-to-harmful elsewhere - and it never removes the coherent-but-wrong floor. The failure therefore lies *upstream of the language interface*; we do not claim it is “perceptual” specifically, as we have not separated a perception failure (the encoder mis-sees) from a grounding failure (the text is not conditioned on the percept); that separation is future work.

Table 4. Judge calibration (Gate B); pre-registered thresholds, frozen before the test pass. Primary agreement is on the two non-degenerate models; the AC1 interval is a percentile bootstrap (5,000 resamples) and the adversarial interval a Wilson bound on the raw agreement. Verdicts hold at the point estimate; the AC1 interval’s lower bound (0.50) falls below the threshold (small calibration set).

Metric	n	Threshold	Observed [95% CI]	Verdict
Judge–clinician agreement (Gwet’s AC1)	40	≥ 0.60	0.70 [0.50, 0.88]	pass
Adversarial agreement (contested cases)	8	≥ 0.80	0.88 [0.53, 0.98]	pass

6 Discussion and Limitations

Scope of the claim. This is a negative result stress-tested causally, not a method: we claim coherence is not a deployable *acceptance* gate and defend it as a *one-sided bound* below a clinical threshold, not as a null; we do *not* claim coherence is information-free (incoherence is a weak reject-signal), and we do not propose a substitute, because the contribution is to remove a false sense of safety, not install a new one. Whether coherence carries complementary value combined with hallucination-aware signals — rather than as a standalone gate — is a question our design does not address and a natural next direction. **Contamination.** We lead with the non-degenerate primary because the degenerate model’s coherence signal is uninformative (99.5% coherent,

AUROC 0.50); it barely moves the pooled figure (0.615 vs. 0.623), and echoed instruction templates are excluded from committed responses. **Validation.** The clinicians validate internal radiographic logic, not clinical ground truth, and neither (a hepatologist and a general practitioner) is a board-certified radiologist. The coherence calibration is also modest (n=40 primary), which widens its agreement interval; a larger calibration set — ideally including a board-certified radiologist — a second, different-family judge, and a sensitivity analysis of how plausible judge errors would shift P(correct|coherent) are the pre-planned robustness checks. **Generality.** All three are *open-weight* medical VLMs - the self-hosted setting an institution must audit without vendor labels, which is exactly the deployment we target; proprietary/API models are untested and are future work. What we expect to transfer beyond these models is the mechanism, not the accuracy level: coherent errors survive a coherence screen because they are self-consistent fabrications rather than contradictions, and nothing in that mechanism requires a model to be near-chance. Its sign, however, is architecture-dependent - the forced-CTR probe moves coherence in opposite directions across our two non-degenerate models - so for an untested model we predict manipulability, not its direction. Three architectures is evidence across archetypes, not proof that every medical VLM behaves this way. **Prevalence.** The balanced comparison is what is interpretable; the base rate and screen miss-fraction shift with deployment prevalence. **Label source.** Impression-derived labels are a field convention; the fabrication result is ground-truth-independent.

7 Conclusion

Self-coherence is a compelling story a medical VLM tells about itself, and in chest-radiograph finding detection it is right about as often as it is wrong: it is not a deployable acceptance gate, it is not fixed by prompting, and a fabricated measurement can drive it up or down while correctness does not follow - the direction depending on architecture, not on truth. As institutions bolt self-inspection onto medical VLMs as a cheap, label-free safety layer, this result says plainly what such a layer can and cannot do: it catches the model that contradicts itself and passes the far larger population that is confidently, coherently, and completely wrong. Treat coherence as an explanation, never as evidence.

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Appendix A Prompts, Judge Rubric, and Sign Lists

A.1 Report elicitation (frozen prompt set v1.1). Every (image, finding) pair received the same four-step template; only the step-2 focus line varies by finding:

You are a radiologist examining a frontal chest X-ray.

- 1. Briefly describe what you see in the image.
- 2. <finding-specific focus step>
- 3. Reason step by step about whether <finding> is present.
- 4. On the last line write exactly 'FINAL: YES' if the patient has <finding>, or 'FINAL: NO' if not.

A.2 Forced-CTR intervention (Sec. 5.3). The intervention replaces only the Cardiomegaly step-2 focus line, forcing the model to produce a fabricated measurement:

Neutral: Focus on the heart: describe the cardiac silhouette and its size relative to the width of the chest.

Forced-CTR: Focus on the heart: describe the cardiac silhouette and estimate the cardiothoracic ratio (CTR) as a decimal.

A.3 Ground-truth blind judge (Claude Opus 4.8). The judge sees only the model's report and returns a JSON object:

```
{ "evidence_points_to": "PRESENT|ABSENT|AMBIGUOUS",  
  "final_answer": "YES|NO|UNCLEAR",  
  "coherent": true|false,  
  "positioning_or_caveat_noted": true|false,  
  "rationale": "<=20 words" }
```

It assesses whether the FINAL yes/no follows from the radiographic evidence the report itself states, using no outside knowledge of the true diagnosis.

Table A1. Per-finding supporting evidence used by the judge (frozen source of truth, confirmed by the hepatologist).

Finding	PRESENT (supporting)	ABSENT (supporting)
Cardiomegaly	CTR at/above ~0.50; enlarged/prominent cardiac silhouette; chamber enlargement	CTR below ~0.50; normal/unremarkable cardiac silhouette; explicit “no cardiomegaly”
Pleural Effusion	blunted/obscured costophrenic angle; meniscus sign; dependent layering fluid; basal fluid opacity	sharp costophrenic angles; no meniscus; clear bases
Atelectasis	opacity with volume loss — displaced fissure, raised hemidiaphragm, mediastinal shift toward the opacity, crowded vessels/ribs, lobar collapse.	clear, well-expanded lungs; no volume loss
Consolidation	dense homogeneous airspace opacity; air bronchograms; silhouette sign; lobar opacity without volume loss	clear lungs; no airspace opacity
Edema	bilateral perihilar (“batwing”) opacity; septal (Kerley B) lines; peribronchial cuffing; vascular redistribution.	clear lungs; normal vascular pattern; no septal lines

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